

STE AGATHE DES MONTS, QC, Canada

WMO: 727200

Lat: 46.050N

Lon: 74.280W

Elev: 395

StdP: 96.67

Time Zone: -5.00 (NAE)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.1	(c) -25.6	(d) -32.9	(e) 0.2	(f) -27.7	(g) -30.6	(h) 0.2	(i) -25.0	(j) 11.0	(k) -12.1	(l) 9.7	(m) -10.2	(n) 2.0	(o) 300	(p) 0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	27.5	20.1	26.0	19.3	24.5	18.4	21.6	25.4	20.6	24.0	19.6	23.0	3.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	15.8	23.4	19.4	14.9	22.4	18.4	13.9	21.5	64.8	25.3	61.3	23.9	58.0	23.0	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.0	(b) 7.9	(c) 6.9	(e) -32.3	(f) 30.0	(g) 1.4	(h) 1.6	(i) -33.4	(j) 31.2	(k) -34.2	(l) 32.1	(m) -35.0	(n) 33.0	(o) -36.1	(p) 34.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-12.6	-10.2	-5.3	3.5	10.7	15.0	18.0	16.5	11.7	5.9	-1.4	-9.9
	DBStd	11.91	7.19	6.29	7.08	5.35	4.53	3.89	3.21	3.61	4.13	4.24	5.18	7.08
	HDD10.0	3128	699	567	474	206	46	5	0	2	28	142	342	617
	HDD18.3	5472	958	800	732	446	239	117	45	79	203	387	591	876
	CDD10.0	774	0	0	0	11	69	154	247	203	77	13	1	0
	CDD18.3	78	0	0	0	0	3	16	34	22	3	0	0	0
	CDH23.3	588	0	0	0	7	37	136	261	131	17	0	0	0
	CDH26.7	83	0	0	0	0	3	24	47	8	1	0	0	0
(14) Wind	WSAvg	2.8	3.1	2.9	3.2	3.1	2.8	2.8	2.4	2.2	2.5	2.8	3.3	3.1
(15) Precipitation	PrecAvg	1236	98	86	93	86	93	113	102	113	105	100	122	123
	PrecMax	1584	185	225	138	140	202	225	223	233	168	158	231	196
	PrecMin	1062	39	20	25	29	27	40	37	25	38	57	78	58
	PrecStd	133	37	50	27	31	46	43	42	48	34	29	35	34
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	6.1	14.3	24.2	27.0	29.2	30.1	27.6	25.7	19.9	13.9	9.3
		MCWB	3.7	2.9	9.0	15.6	16.1	19.8	21.8	21.0	19.5	15.5	11.2	8.5
	2%	DB	1.5	3.4	9.1	18.9	23.9	27.0	27.9	26.2	22.8	17.4	11.2	4.1
		MCWB	0.6	1.5	6.1	12.1	15.7	19.1	21.0	19.8	18.3	13.9	10.0	3.4
	5%	DB	0.0	1.3	6.3	15.3	21.6	24.7	26.2	24.9	21.0	15.0	8.8	1.4
		MCWB	-0.4	0.0	3.4	9.1	14.4	17.9	19.8	19.2	17.6	12.1	7.7	0.4
	10%	DB	-1.8	-0.2	3.7	12.2	19.3	22.5	24.5	23.3	19.1	12.9	6.0	-0.3
		MCWB	-2.4	-1.1	1.6	7.3	13.2	16.3	18.5	18.7	16.0	10.3	5.0	-0.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	3.9	10.4	16.2	19.7	21.5	23.3	22.8	20.7	16.7	12.1	8.8
		MCDB	4.1	5.6	12.1	22.7	22.7	26.4	28.2	26.2	23.2	18.8	13.4	9.3
	2%	WB	0.8	1.6	6.7	13.3	17.3	20.3	22.0	21.2	19.5	14.9	10.1	3.6
		MCDB	1.3	2.7	8.9	17.4	20.7	24.9	26.7	24.1	21.8	16.6	11.0	4.0
	5%	WB	-0.5	0.3	4.0	10.1	15.9	19.1	20.9	20.2	18.0	12.9	7.5	0.6
		MCDB	0.0	1.2	5.6	13.8	20.1	23.8	24.3	23.2	20.5	14.5	8.1	1.2
	10%	WB	-2.4	-1.3	1.7	7.8	14.4	17.7	19.8	19.2	16.5	10.9	5.1	-0.8
		MCDB	-1.7	-0.1	3.3	11.4	18.3	21.1	23.3	22.2	18.7	12.6	6.2	-0.3
(35) Mean Daily Temperature Range	MDBR		10.0	10.3	10.6	9.8	11.4	10.8	10.7	10.1	9.9	8.7	7.1	9.1
	5% DB	MCDBR	10.6	8.9	11.6	15.4	14.9	13.2	12.5	11.6	11.9	11.8	11.0	10.2
		MCWBR	10.5	7.8	8.4	9.2	7.8	6.6	6.1	5.8	7.5	8.2	10.2	10.1
	5% WB	MCDBR	10.4	8.4	10.7	14.1	12.0	11.5	10.4	9.4	10.9	10.0	10.2	10.1
		MCWBR	10.5	7.9	8.6	9.3	7.1	6.6	5.9	5.3	7.7	8.2	10.2	10.1
(40) Clear-Sky Solar Irradiance	taub		0.243	0.269	0.303	0.328	0.357	0.400	0.389	0.379	0.348	0.319	0.291	0.261
	taud		2.354	2.263	2.269	2.424	2.413	2.286	2.323	2.356	2.438	2.532	2.515	2.363
	Ebn at Noon		879	912	925	932	910	867	872	868	869	851	816	808
	Edh at Noon		72	97	114	108	113	129	123	114	96	76	63	65
(44) All-Sky Solar Radiation	RadAvg		1.37	2.33	3.68	4.55	5.13	5.60	5.59	4.93	3.84	2.24	1.34	0.98
	RadStd		0.19	0.28	0.34	0.42	0.51	0.41	0.50	0.24	0.33	0.21	0.17	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page